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# Implications of offshore wind energy developments in coastal and maritime tourism and recreation areas: An analytical overview

Júlia Terra M. Machado [a,*], Maria de Andrés [b]

[a] School of Political Science & Sociology, National University of Ireland Galway, Galway City H91 TK33, Ireland
[b] Research Group in Integrated Coastal Zone Management, University of Cadiz, Spain

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ABSTRACT

Offshore Wind Farms (OWFs) became promising investments to meet climate change targets while bringing autonomy and development to several coastal countries. However, these developments may cause externalities, primarily affecting users dependent on maritime space and its seascapes, such as Tourism and Recreation (T&R). The marine renewable expansion, in general, is expected by some of the actors involved to affect T&R with the prospect of decreasing local welfare and the number of visitors. Nonetheless, this influence over Coastal and Maritime Tourism & Recreation is not well addressed by academic papers. Here we assess this literature gap via a semi-systematic narrative meta-synthesis (SSNMS) which expands on the methodology and results of two guiding reference studies (Scottish Government, 2008; Parsons & Firestone, 2018) to conjecture about this topic. The SSNMS promotes the evaluation studies produced between 2015 and 2019 selected through a rigorous investigation of relevant peer-reviewed keywords in Scopus and Web of Science. The analysis revealed that the studied knowledge gap persists and that OWFs can indeed impact coastal communities dependent on maritime tourism and recreational activities, despite the obligatory Environmental Impact Assessment for OWFs. However, both negative and positive impacts greatly vary with development stages and locations. Nevertheless, results still provide valuable conclusions common to generic coastal areas dealing with offshore wind farms sitting. In the few cases where authors consider T&R in their impact analysis, the results indicate a lack of public sensitivity and participation. Higher clarity and participation in the process might play a key role in generating optimistic attitudes toward OWF projects and positive synergies in between the local Blue Economy actors. In conclusion, the projected negative impacts mainly emerge as a reaction to the absence of proper integration between local T&R stakeholders and OWF developers, demonstrating that adverse effects are mitigable and preventable. Furthermore, the results indicate that a distinctive Tourism & Recreation Impact Assessment should be part of the obligatory technical assessments submitted by the OWF developers in coastal areas that rely on these activities.

## 1. Introduction and objectives

Globally, coastal cities and agglomerations' growth trend keeps increasing, even more intensely in emerging countries (Barragán and de Andrés, 2015). Approximately two-thirds of Earth's population lives within 60 km of the coast. Almost half of the world's cities with more than one million inhabitants are located in estuaries, and a significant number of countries have more than 70% of their population living within 100 km of the coast (Fig. 1). This situation projects a scenario of inadequate urban development, inappropriate wastewater treatment, and inadequate means to protect the quality of coastal waters, increasing energy demands and greenhouse gas emissions. Nevertheless, this consumption and population growth around coastal regions, and the increasing intensity of climate change effects, urged the development of several international actions to reduce $CO_2$ emissions, including massive efforts toward transitioning to a cleaner energy matrix.

The coastal environment is a leader in ecosystem services and contributions to the global economy, but only healthy marine ecosystems increase resilience to climate change impacts and secure the sustainable development of activities such as energy undertakings, maritime transport, tourism, fisheries, and aquaculture. An integrative maritime economy assists these sectors in accomplishing environmental goals and nature conservation (e.g., genetic diversity) without jeopardizing the production of goods, life-support processes (e.g., water purification), or

* Corresponding author.
*E-mail addresses:* j.mirandamachado1@nuigalway.ie (J.T.M. Machado), maria.deandres@uca.es (M. de Andrés).

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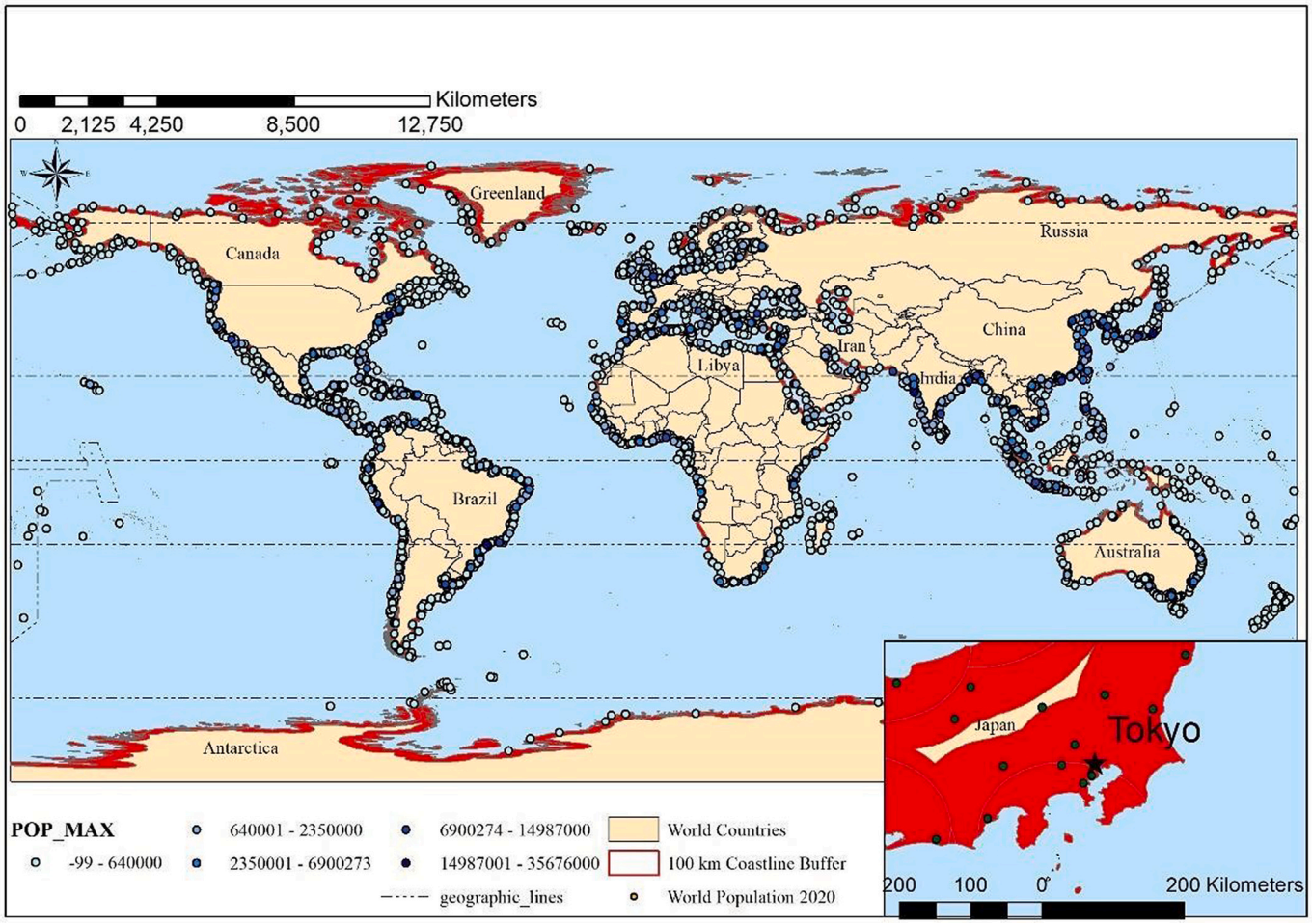


**Fig. 1.** Cities within the 100 km buffer distance from the coastline. [Data source: Natural Earth open-source database].

life-fulfilling conditions (e.g., beauty, and recreation opportunities) (Van den Burg et al., 2019). However, the same sectors can create ecosystem pressures on, for example, food security and safety, affecting the net economic value of tangible and intangible marine ecosystem services directly and indirectly. This behavior is particularly evident in the Tourism & Recreation and Offshore Wind sectors.

On one side of this discussion, we have influential offshore wind developers finding their place in the energy market. On the other side, we have several actors responsible for the coastal and maritime Leisure Industry (tourism and recreation).

The first sector, wind energy, has had record installations worldwide, driven by the constant political and technological development toward reaching the 2050 climate targets, with China planning to build 30GW Offshore Wind Farms by 2030 and Europe breaking both onshore and offshore installation records. The Global Wind Energy Council (GWEC, 2019) states that Europe had nearly 84% (15,780 MW) of all offshore installations concentrated in the waters off the coast of eleven European countries, including 7 of the worldwide top10 countries in cumulative Offshore wind capacity in 2017: the UK, Germany, China, Denmark, Netherlands, Belgium, Sweden, Vietnam, Finland, Japan. These OWF installations have a 25–30 years lifetime, whereas development time is between 7 and 10 years (European MSP Platform, 2018), and their structures depend on local characteristics such as wind intensity, water depth, and distance from the shore.

The former, the tourism and recreation sectors combined, is considered a crucial sector and source of revenue, with over 13 million people in the European Union (EU) employed in economic activities related to tourism in 2017 (European Union, 2019). According to EUROSTAT (Eurostat, 2020a), European tourism has coastal regions as the most popular destinations for domestic and international tourists in several cities (Fig. 2). However, these leisure-related activities[1] are part of a complex management system, which continuity depends on the quantity and frequency of visitors (European MSP Platform, 2018). Moreover, the need for particular environmental and infrastructural conditions significantly characterizes coastal tourism or beach recreational activities, which makes them highly reliant on the maintenance and protection of these characteristics.

The literature on wind energy externalities and tourism and recreation alone is extensive, with holistic quantitative and qualitative studies about management and impacts. Several research projects have been investigating the development of Marine Renewable Energy (MRE), which sadly does not reflect the number of analyses covering the effects on Tourism and Recreation (T&R). However, the same does not seem to apply to analyzing the combination of these sectors. Therefore, there is a need to compile and analyze the studies that have been carried out on the perception of the impacts of OWFs on maritime areas with different levels of T&R.

This research, then, is performed to identify potential information and literature gaps on the possible effects created by the interaction between the two sectors while understanding its mutually influencing nature. For that, the results of two pioneering works, "The economic impacts of wind farms on Scottish tourism" (Scottish Government T,

[1] Here, recreation falls under the definition of leisure, commonly defined as part of the Sports & Recreation market for data analysis.

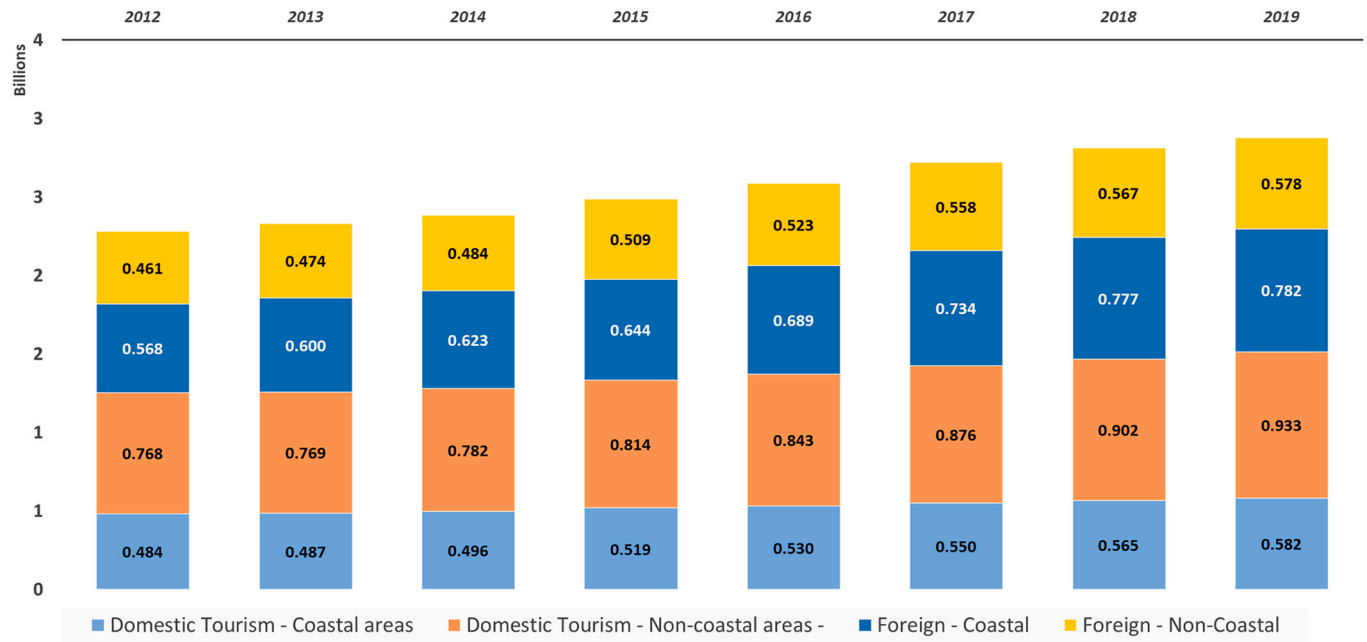


**Fig. 2.** Nights spent at tourist accommodation establishments* by coastal and non-coastal areas from 2012 onwards (Eurostat, 2020b).
* Hotels; holiday and other short-stay accommodations; camping grounds, recreational vehicle parks, and trailer parks.

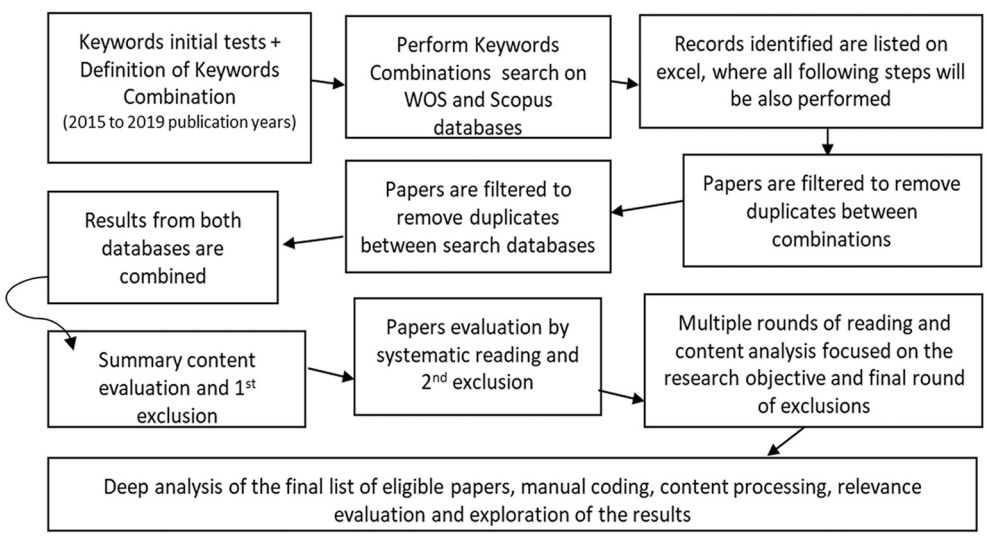


**Fig. 3.** A schematic diagram detailing the process of the semi-systematic narrative meta-synthesis (SSNMS) performed.

2008) and "Atlantic Offshore Wind Energy Development: Values and Implications for Recreation and Tourism" (Parsons and Firestone, 2018), assisted in developing the research objectives of the present study. The first presents, on a general analysis, that landscape interruptions inevitably affect tourism, resuming the studied relationship on a scale of the extent of impact and not of whether it exists or not. The second, written a decade later, raised again this impact hypothesis, covering the US coast from Massachusetts to South Carolina. However, this time, the authors assume that OWF projects may affect coastal tourism and beach use if they alter the seascape, and their findings focus on the impact on major tourist destinations, basing their analysis on famous cases of conflict.

Based on the results and gaps found in these two papers, six hypotheses and three main objectives were formulated to guide this overview, as detailed in the methodology session. The three main objectives of this study are:

- Uncover the literature produced on the existence of impacts generated by the siting of Offshore Wind Farms (OWF) on Tourism and Recreation areas, investigating its different effects on the tourism level, OWF project's phase, geomorphological factors, turbine-driven landscape interruptions, as well as the persistence and stability of impacts generated;
- Analyze the OWF's acceptance among the T&R stakeholders to be impacted or already impacted by an energy park, having the literature overview to define which parameters are key in altering acceptance behaviors among this set of coastal inhabitants and visitors;

- Discuss the importance and format of the Tourism & Recreation Impact Assessment.[2]

## 2. Method

Here we intend to study the possible impacts generated by the siting of offshore wind farms (OWFs) in areas characterized by tourism and leisure activities via reviewing the relevant publications. This review is performed using a modified structure based on the methodology developed and performed by the report "The economic impacts of wind farms on Scottish tourism" (Scottish Government T, 2008). This time, the positive and negative impacts investigated here support the discussion via critically reviewing cases of alterations in visit/stay numbers and recreation-driven income and behavioral changes, and via establishing the stakeholders involved and the different impacts associated with the multiple stages of developing an OWF. The six guiding hypotheses, employed here to organize the literature overview and the points for discussion are:

- Hypothesis 1OWFs generate impacts on Tourism and Recreation, and the negative impact of OWFs on tourism at the national level is small, with non-significant job loss;
- Hypothesis 2OWF Landscape interruptions inevitably bring tourism and recreation impacts, reducing tourist demand and leading to significant economic losses;
- Hypothesis 3Communication is a key parameter to increase acceptance among tourism stakeholders;
- Hypothesis 4Green marketing is a key parameter in growing acceptance among tourism stakeholders;
- Hypothesis 5Tourism Impact Statement must be part of the Environmental Impact Analysis of OWF;
- Hypothesis 6OWF's impact on coastal and maritime (C&M) Tourism and Recreation changes through time and with project status/phase.

However, although also multidisciplinary and having no geographical limitations, the methodology used for this evaluation differs from the Scottish study in three key points. First, contrary to previous works, this overview includes studies that focus on the recreation perspective separately rather than having its inputs eclipsed and generalized by the records on tourism. Second, it expands and complements the structure of the study by following the rules for systematic reviews defined by Siddaway et al. (2019) and Snyder (2019) for semi-systematic analysis. Third, by analyzing papers published from 2015 to 2019, we follow a crucial moment in the development of the offshore wind sector worldwide, a transitioning moment, with records of new production capacity. Nevertheless, by expanding the base methodology, we offer a more comprehensive range of results in a more restricted study regarding its focus on the offshore environment. Thus, the present work builds a semi-systematic narrative meta-synthesis (SSNMS), which follows the steps in Fig. 3.

Following these definitions, we establish the progress of the hypothesized problem by identifying relations, contradictions, gaps, and inconsistencies. Thus, the assessment is performed as a critical overview of the literature, where we explore the formulated hypotheses and discuss them under three main subheadings:

- Indicators and impacts;
- Information, Public Acceptance, and Stakeholders' engagement;
- T&R Impact Assessment.

The exploration of international studies about Offshore Wind Farms and their impact on coastal Tourism was performed on electronic databases, using a combination of keywords. First, the research question is defined and fragmented into keywords. The combinations (Table 1) were selected based on their relevance, on an evaluation of the frequency in which these particular keywords appeared in a diverse range of related news, articles, and white papers discussing (separately) both sectors here analyzed. Next, we searched the combinations in Web of Science and Scopus, two online databases of relevant peer-reviewed academic articles, to ensure a replicable, precise, and trustworthy assessment.

The selection of the relevant articles involved, first, compiling the findings. Then the triage phase started with the exclusion phases. The criteria adopted were guaranteeing the paper's ability to suit at least one of the hypotheses defined and assuring it discusses offshore wind farms rather than only onshore parks. Then, in a second exclusion phase, the following inclusion and exclusion criteria were implemented:

- The articles must address, directly or indirectly, the views of tourists and recreationists, both domestic and international;
- No geographic area or field/theme/sub-theme is specified;
- The publication period includes only papers published between 2015 and the end of 2019.
- Papers with study cases and surveys or other sources of direct public opinion are preferred (relevance filter);
- The articles must be aligned with the refinement method developed by the authors (relevance filter).
  - To increase the degree of refinement, a relevance scale monitors the quality of the scientific article found. In this context, some words more directly connected to the title of this work and its main goals, such as "Visitor", "Tourist", "Recreation" and its variations, and "Offshore" became indicators for relevance. Without acting as an exclusion parameter for most cases, this relevance analysis illustrates the publications that focus on tourism and recreation.

Examining the peer-reviewed articles in detail, through consecutive guided readings, was also part of the method. Our methodology considers how innovative this topic is, and how unclear and ambiguous the findings could be, as positive, negative, and neutral results were

**Table 1**
List of the searched combination of keywords.

| Keywords (combination) WOS and Scopus | |
|---|---|
| Offshore wind farms impact coastal tourism | Offshore wind coastal recreation |
| Offshore wind public opinion tourism | Offshore wind conflicts Tourism |
| Offshore wind turbines tourism | Offshore wind public perception |
| Offshore wind landscape impact | Offshore wind sports activities |
| Coastal tourism renewable energy | |

[2] Tourist Impact Assessments (or Statements) are studies elaborated by OWF developers about the likely impacts due to the development of OWFs on the local tourism industry and the methods that can minimize costs and negative impacts while maximizing benefits. "The length of such statements will inevitably be dependent upon the importance of tourism in the local area." (Scottish Government, 2008, pp269).

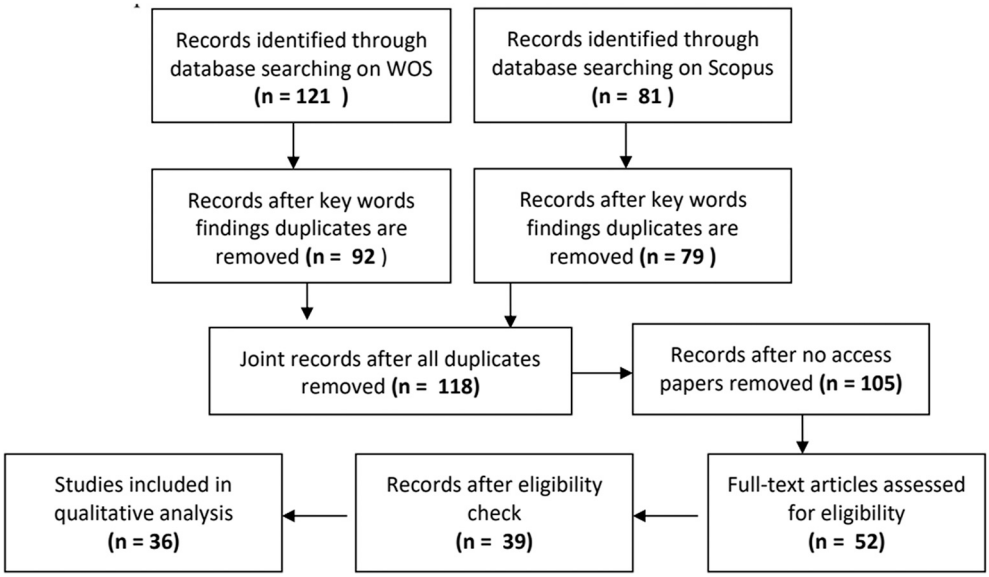


**Fig. 4.** Methodology for the selection process explained – flow diagram.

expected. Hence, to compile these results in a binary model, neutral results were converted into positive and negative results after the authors evaluated these papers, based on their stronger tendencies. Consequently, the classification of positive and negative impacts for this overview follows the following definitions:

- Positive impact: the study shows positive or mainly positive impacts obtained from the interaction between OWF and Tourism and recreation sector or from landscape impact justified beneficial to these sectors;
- Negative impact: the study presents negative or mainly adverse effects obtained from the interaction between OWF and Tourism and recreation sector or from landscape impact justified as harmful to these sectors.

## 3. Results

### 3.1. Search process results

Following the previously explained methodology, 202 studies were found and refined through a first exclusion round based on repetition between the databases. This first exclusion provided 118 peer-reviewed scientific papers that moved to the next phase. After all the exclusion phases, 36 eligible works continued to the qualitative analysis (Fig. 4). By considering the different spatial, socio-cultural, and political settings, the selected works identify and assess the recurrent processes or causalities that lead to the possible impacts.

Based on the number of crosschecking stages and repeated references, the 36 papers collected have shown significant relevance to the objectives. Most papers (53%) were present in both platforms (WOS and Scopus), while 11% only appeared in Scopus and 36% only in WOS. The low number of relevant papers, specifically addressing the renewable energy impacts through the lenses of recreation and tourism sectors, meets the authors' expectations and the findings of other authors (Brownlee et al., 2015; Kipperberg et al., 2019).

The radar chart below (Fig. 5) details the findings of our extra significance analysis, performed to improve the confidence of the results, as defined in our methods section. The relevance ranges from 0 to 8. All but one of the selected 36 published papers complies with at least one of the defined conditions, confirming the suitability of the articles regarding their significance toward the hypotheses-focused results and discussion. Nonetheless, the study that scored zero due to its more comprehensive nature and unspecific language regarding renewable energy infrastructure projects is still applicable. Although this filter presented a consistent way to characterize the papers beyond following the searched keywords, this last filter is, to the authors, one of the tools to be improved for future works. The numbers in this chart are the codes used to identify the papers analyzed, and the complete table with the values and significance is part of the appendices section.

### 3.2. Semi-systematic narrative meta-synthesis (SSNMS) results

The 36 peer-reviewed papers obtained (See Table 2) after performing the semi-systematic narrative meta-synthesis (SSNMS) provide a wide range of answers, opinions, and experiences via discussions, technical analysis, and study cases. Ninety-two percent of the studies indicate that siting offshore wind farms in coastal areas impacts coastal tourism and recreation activities.[3] In other words, the articles, strongly, point to a relationship of influence between the sectors analyzed when they coexist. Adding to the negative impact results, most neutral results had negative inclinations after they were recategorized according to their tendency toward positive or negative impacts. Therefore, among the 36 papers analyzed, 68% find negative impacts or expect the impact to be negative (Fig. 6), confirming our hypothesis.

The different authors discuss cases of implementation successes, which mention multi-use and synergies, and implementation failures, where installation of these structures has been suspended or delayed. Included in this list of international cases, the studies contain discussions about the Block Island OWF (Firestone et al., 2018a; Sokoloski et al., 2018; Griffin et al., 2015; Smith et al., 2018), Goto city project (Shyam Kularathna and Takagi, 2018), Jeju Island wind power complex (Kim et al., 2016), Cape Wind project (Brownlee et al., 2015; Hall and Lazarus, 2015; Bush and Hoagland, 2016; Gudding et al., 2018), Maine's DeepCwind Consortium (Hall and Lazarus, 2015), Walney OWF and other OWFs based in the UK (Rydin et al., 2018), Inch Cape Wind farm (Martino et al., 2019). Regarding the location of the studies, most authors analyzed communities or regions bathed by the Atlantic Ocean and studied areas in Europe (43%) and the United States (41%). By far,

[3] In the different methodologies and parameters used, the *Yes* response describes articles in which the influence of OWF is felt by the stakeholders involved and by the sector, while the *No* response describes articles where this influence was considered non-existent or irrelevant.

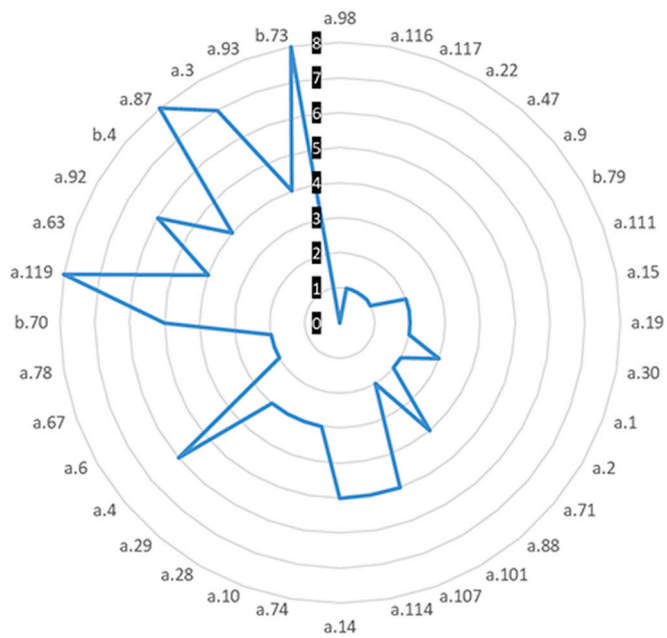


**Fig. 5.** Radar chart analyzing the scientific papers by relevance factor.

**Table 2**
Summary of the results and their categories (impact existence and type of impact).

| Code | Year | Impact | Study response |
|---|---|---|---|
| b.73 | 2015 | Yes | Negative |
| b.4 | 2015 | Yes | Negative |
| a.2 | 2015 | Yes | Negative |
| b.79 | 2015 | Yes | Neutral |
| b.70 | 2015 | Yes | Negative |
| a.3 | 2015 | Yes | Positive |
| a.1 | 2015 | Yes | Neutral |
| a.4 | 2015 | Yes | Positive |
| a.14 | 2016 | Yes | Negative |
| a.6 | 2016 | No | |
| a.10 | 2016 | Yes | Negative |
| a.9 | 2016 | Yes | Negative |
| a.15 | 2016 | Yes | Negative |
| a.19 | 2016 | Yes | Negative |
| a.30 | 2017 | No | |
| a.22 | 2017 | Yes | Negative |
| a.29 | 2017 | Yes | Negative |
| a.28 | 2017 | Yes | Negative |
| a.92 | 2018 | Yes | Negative |
| a.88 | 2018 | Yes | Positive |
| a.98 | 2018 | Yes | Negative |
| a.67 | 2018 | Yes | Positive |
| a.87 | 2018 | No | |
| a.93 | 2018 | Yes | Positive |
| a.71 | 2018 | Yes | Neutral |
| a.78 | 2018 | Yes | Negative |
| a.63 | 2018 | Yes | Positive |
| a.47 | 2018 | Yes | Neutral |
| a.74 | 2018 | Yes | Positive |
| a.117 | 2019 | Yes | Negative |
| a.119 | 2019 | Yes | Neutral |
| a.107 | 2019 | Yes | Negative |
| a.116 | 2019 | Yes | Neutral |
| a.101 | 2019 | Yes | Positive |
| a.114 | 2019 | Yes | Negative |
| a.111 | 2019 | Yes | Negative |
| TOTAL | 36 | | |

although the number of developments and their size are far from comparable with the European installations, the US is the country that performs or allocates most of the relevant studies.

Regarding the No impact studies, the investigation of OWFs in the Netherlands (Nordman et al., 2016) concludes that no change in beach-going plans happened because of the wind farm. This work also shows an increasing percentage of "would visit the beach just as often (or more)" from the pre-installation phase to one year of operation, 81% for tourists and 82% for recreationists. The majority of participants questioned by the studies reviewed think that recreational boating would either benefit from or not be affected by offshore wind parks. Yet, demographic factors can greatly affect this statement and will be discussed further.

Offshore structures for renewable energy generation can alter not only the landscape but the area access for several other maritime and coastal users, including shipping, fishing, tourism, and recreational users (Alexander et al., 2016). Safety was also discussed in the papers and, although some papers commented on potential risks involving OWF Power Cables and Recreational fishing and boating, most authors would have an overall response of No Impact for safety-related impacts (Sijmons et al., 2017; Firestone et al., 2018b).

Regarding Synergies, among the results, 2 papers study the connection between OWF and Tourism and Recreation through a Multi-use (MU) perspective (**Depellegrin et al., 2019**; **Calado et al., 2019**). These do not show OWF and Tourism directly combined but discuss how they can be part of the maritime MU scenario in various combinations with other marine users in combinations such as Fishery & Tourism & Environmental Protection, Aquaculture & Tourism, Floating Offshore Wind (FOW) farms & Aquaculture, FOW farms & Environmental Protection, and OWFs & Aquaculture. The success of these MUs is grounded on how tourism-driven MUs have the highest potential for successful synergies, foreseeing a successful combination between OWF and Tourism.

In the articles where the existence of impact is defined, it is possible to notice that surveys with groups clustered by age provide well-defined and opposite outcomes. The impact on tourism and recreation appears intrinsically connected to local traditions, culture, socio-economic, and demographic factors such as age, education, and income (Firestone et al., 2018a; Sokoloski et al., 2018; Smith et al., 2018; Lin et al., 2019). For example, experienced fishers and aquaculture farmers tend to have high opposition or neutral positions about the OWF, while the younger and more educated workers have higher approval rates. Westerberg et al. (2015) find that socio-demographic factors are consistent and do correlate positively with age and income, whilst the evidence of the impact of education on preferences is mixed, with higher educated respondents, on average, being less sensitive to the installation of Offshore Wind Farms. Chen et al. (2015), mainly studying fishery and aquaculture actors in Taiwan, find that lower age, lower experience, and higher educational level respondents would provide higher approval rates of OWFs and focus on the benefits of OWFs. Additionally, Brownlee et al. (2015) find that male respondents are generally less supportive of offshore wind developments, as female respondents demonstrate being more affected by landscape and visual impact, which is a result also discussed, in higher or lower degrees, by other authors. Nevertheless, behavior predictions using demographic factors are not black-and-white parameters. The case of Block Island and Rhode Island (Firestone et al., 2018a), for example, demonstrate how front neighbors and deeply connected communities with similar incomes and education can have different levels of place attachment, which is also described in Brownlee et al. (2015) for South Carolina's communities (USA), and by Bates & Firestone (2015) for Atlantic City and Delaware (USA).

Throughout the results, most frequent visitors interviewed showed no significant promptness to any sort of instant change of behavior in the

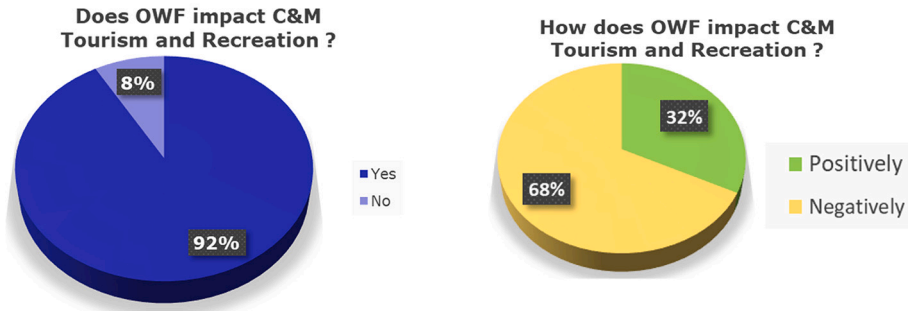


**Fig. 6.** Findings in percentage: Existence of impact (Barragán and de Andrés, 2015) and binary type of impact (Van den Burg et al., 2019) generated by the coexistence of an OWF and C&M Tourism/Recreation activities.

frequency of their visits due to the installation of OWF, although they do believe that OWF detracts from the visitor experience. Shyam Kularathna & Takagi (2018), with a very comprehensive set of questions, find that respondents expect positive impacts on job creation (78%), tourism-related businesses (69%), and property values (46%). Lin et al. (2019) focus on a scenario where Integration between the OWF and the local stakeholders is perceived as an opportunity to provide green energy for residents and industries while also providing a basis for new educational energy tours and DIY courses for students and tourists. Nonetheless, the articles reveal that the public opposition is mainly defined by possible issues driven by visual disamenities and by how OWFs "Decrease scenic and natural beauty" and "Negatively influence the marine environment". On the other hand, public support appears driven by how OWFs will "Benefit future generations" and "control climate change" by producing clean electricity. Most supporters also believe in the benefit of OWF, in its prospect to develop multiple marine recreational activities (Shyam Kularathna and Takagi, 2018; Rudolph et al., 2018).

Regarding physical and structural parameters, all results show higher levels of rejection closer the projects are to the shore, but the definition of close varies. Some articles discuss that OWF developments should be kept between 15 and 20 km from shore (Nichifor, 2016; Christoforaki and Tsoutsos, 2017; Bonar et al., 2015; Depellegrin, 2016), and other papers define further distances of 30 km from the coast (Firestone et al., 2018a; Griffin et al., 2015; Westerberg et al., 2015; Nichifor, 2016; Voltaire et al., 2017). The study about Catalonia (Voltaire et al., 2017), in Spain, concludes that wind farms can be major visual disruptions at distances up to 13 km, assuming the existence of negative visual impacts when any OWF is less than 24 km from the coastline. Another study, performed in the region of Languedoc Roussillon, in France, (Westerberg et al., 2015) concludes that local environmental-associated objections to placing a turbine 12 km offshore could get disregarded based on the environmental gains provided, which remove the disutility of the windfarm placement even for those who consider wind power to be costly to society. Another interesting finding is the discussion over the effect of the structures on tourism and visual pollution as a cross-boundary impact mutually affecting neighboring countries. Nichifor (2016) shows how two neighboring countries, Romania and the Netherlands, can have different opinions on the matter, with only 4% of the Romanian respondents and 17% of the Dutch respondents not seeing any significant visual impact.

Most oppositions regard the possibility of harming the seascape (Firestone et al., 2018b; Nordman et al., 2015). Nordman et al. (2015) find that, generally, more developed seascapes are considered more acceptable sites for offshore wind farms than more natural areas. Christoforaki & Tsoutsos (2017) state that sites with key productive tertiary activities (such as tourist/recreation areas), sites in which the landscape is a determinant factor for the activities, should be rejected as OWF sites. C. K. Kim et al. (2018) state that offshore wind farms reduce the tourism value, which leads to local economic vulnerability, and should avoid areas contain Marine National Parks, Natural Environmental Conservation Zones, and Ecosystem and Landscape Conservation Zones, besides many beautiful islands that provide recreation and aesthetic services. The density of turbines, the beach size, and morphological characteristics can also lead to reductions in the demand for beach visits. For example, Voltaire et al. (2017) estimate a significant welfare loss to Catalonia's beach visitors, between 67.3 and 203 million euros per season, if the theoretical 570 MW offshore wind power to be generated along the Catalan coast gets installed. Johansen (2019) finds that second-house owners see the impact on tourism as the third higher anticipated negative effect from the siting of OWFs when compared to coastal landscape, noise levels, marine life, flora & fauna, local jobs, local economy, and "my use of local area". Klain et al. (2018) find that recreation and tourism aesthetic-related activities are not expected to benefit from offshore wind farms, but New Zealand participants expect recreational fishing to benefit from the artificial reef associated with the bases of the turbines. Martino et al. (2019) analyzed several countries and concludes that impacts on tourism and recreation are mainly caused by the top-down approach, lack of attention to local stakeholders, and the non-participatory planning process (or low quality of the ones developed).

Other topics that seem to attract significant attention in this discussion are Islands and the Cape Wind Project in the USA. Authors de Groot & Bailey (2016) discuss how islands have high rates of acceptance among their visitors and (seasonal or full-time) residents despite having the tourism and recreation industry as their main source of revenue and being a particularly attractive location for second homeowners and retirees. Smith et al. (2018) complement this analysis by discussing how traditions and recurrent visitors are deeply connected to small communities and the usual geographical concentration of recreation and tourism activities. These characteristics of islands make the findings in these areas more direct and bring key findings that are likely to be extrapolated to other scenarios, as islands are great test beds for innovation. Nevertheless, despite Islanders being indeed more sensitive to projects and changes, their issues with energy security due to their high dependence on fossil fuels are likely to make them more inclined to support renewable energy projects than near-ocean coastal respondents (Firestone et al., 2018a; Gudding et al., 2018).

A third and final important topic mentioned by the authors analyzed, which is directly or indirectly discussed by the papers, is the methodology used in the evaluations of impacts, and the quality of the tools and instruments used in the materials provided by OWF developers and by the researchers. The significance of landscape-related impacts shows the importance of using quality tools for inspecting the visual effects of these developments to lower levels of uncertainty. Thus, more dynamic and realistic representations can facilitate understanding the real and the perceived distance between the turbines and the tourists and recreationists (Teisl et al., 2018). However, throughout the results, the improvement of the representation connects to an increase in the number of negative opinions on how a wind farm will affect their experience and visitation.

## 4. Discussion

The results provided a thorough evaluation of scientific papers, including their main results and conclusions, and best practices in different countries to build a discussion around the possible impacts of the siting of offshore wind turbines in coastal areas with recreational and tourist activities. These findings become indicators of the analyzed sectors' relationship and the degree to which the literature gap is currently analyzed, transforming this review into an indirect Impact Analysis (IA).

The study confirms the existence of the conjectured literature gap and demonstrates the persistence of this gap, which represents a critical issue for offshore wind farms and coastal and maritime tourism & recreation businesses. Most studies referred to in this review suggest no evidence of a meaningful adverse economic impact produced by the construction of offshore wind farms in the selected niche. The studies also reveal that the most common answer given by the relevant actors is the expectation of adverse effects, which leads to the overall statement of possible negative impacts in the majority of scientific articles. On the other hand, there were a few cases in which wind turbines caused visiting losses or gains in particular regions, positive and negative changes in beach attitude, and some direct and indirect adverse impacts on local revenues.

Here, in light of the results obtained through the detailed analysis of the 36 papers, we have raised a significant discussion over the several possibilities of impacts and solutions for the analyzed ecosystem. First, the relationship between the sectors when they coexist in coastal areas shows that the impacts were uncovered by analyzing its properties regarding tourism level, OWF project's phase, geomorphological factors, turbine-driven landscape interruptions, and the persistence and stability of effects generated. Second, socioeconomic aspects such as communication and sustainability drive acceptance as indicators of behavior and course prognosticators. Lastly, we discuss the need for a Tourism Impact Statement based on the quantitative and qualitative results obtained through this semi-systematic narrative meta-synthesis.

### 4.1. Indicators and impacts

#### 4.1.1. Place attachment and protective responses

People can feel varying degrees of belonging and attachment, ranging from the neighborhood to the regional, national, and even global levels (Firestone et al., 2018a; Johansen, 2019; Devine-Wright, 2009). In general, this place attachment feeling relates to natural beauty, pristineness, ecological integrity, exploration, recreational enjoyment, and economic development (Firestone et al., 2018a).

The 'NIMBY' (Not In My Back Yard) concept is commonly used to explain this sort of attachment-driven public opposition to new developments near homes and communities, occurring particularly with wind farms or electricity pylons (Devine-Wright, 2009). The attachment found is suggested to motivate the delay or even cancelation of many local projects, despite the general acceptance of OWF (Bates and Firestone, 2015). Although this NIMBY syndrome (or Nimbyism) does not commonly seem to affect tourists (as they do not live in the area), recent studies show that NIMBY can also appear related to a strong feeling of place meaning as tourists often seem attached to local's natural settings. This behavior is even more likely to influence recurrent visitors such as recreationists, as this concept contemplates both tourists and locals, automatically carrying a stronger connection with place-protective feelings. Additionally, NIMBY effects can also be a protective behavior. The "Backyard" is extended to a community area, in which the sector of interest or affection can be impacted and create opposition. Cities with higher shares of income provided by T&R are more likely to be worried about its impacts, leading to strong opposition to the OWF projects and a consequent decrease in interest in the area by the OWF developers (Sokoloski et al., 2018; Johansen, 2019). This extended NIMBY concept brings us closer to place attachment effects.

Identifying place identity, place dependence, and community social attachment are parameters exhaustively mentioned in articles related to OWF. These seem to be vital points to consider when developing analyses, reports, and public consultations, including local integrity and heritage (Kermagoret et al., 2016). But activities in which recreationists and tourists are involved, as well as the type of recreationist and tourist, also seem to affect acceptance and opposition rates.

#### 4.1.2. Do physical parameters matter?

Multiple physical and non-physical parameters can affect a population based on the spatial challenges arising from considering community benefits for offshore developments: the distance between the location of a project and any beneficiaries, the difficulties of defining the community, and how the impact is perceived (Rudolph et al., 2018). The list of studies suggests that large structures, such as wind turbines, can be considered either a symbol of progress or an intangible loss. These possible visual disamenities caused by the OWFs evaluation are calculated and demonstrated in several works as an indicator for impact analysis (Kipperberg et al., 2019; Westerberg et al., 2015; Voltaire et al., 2017; Lamy and Azevedo, 2018).

Although these so-called impressive representations of industrial and technological advances, wind turbines, and their possible impacts on visual amenities permanently became part of environmental impact assessments, it took decades for appropriate technologic solutions to reach the market. The disturbance of the visual landscape resources caused by Offshore Wind Farms, according to the authors analyzed, tends to decline with increasing distance from land (Griffin et al., 2015; Nichifor, 2016; Kim et al., 2018; Teisl et al., 2018). Between the results, a 6-mile distance between the offshore wind farm and the coast is the minimum distance to lessen aesthetic impacts (Nordman et al., 2015). Consequently, developers wish to explore options for beyond-horizon offshore activities, far from tourist destinations and coastal communities. They pushed for constant technological advances in Marine Renewables, leading to changes in the size and shape of the blades and, for example, the introduction of floating structures (Hall and Lazarus, 2015). Although most wind farms are bottom-fixed (WindEurope., 2020), the number of developers exploring deeper waters and stronger winds with floating structures is likely to significantly increase as the technical limitations decrease. Therefore, expanding OWF toward locations farther from shore can prevent potential adverse landscape effects, but it could also pose barriers to several blue economy multi-uses of this space.

Nevertheless, distance does not only concern visual impacts, as it is an imperative parameter for increases in electricity transmission costs; choosing the location seems to be a matter of combined suitability, a fair balance between low visibility impact and OWF siting and operation costs. Likewise, local perceptions can be highly connected to the development profile of coastal cities, and the existing versus expected development facade. The allocation of turbines in heavily developed coastlines with industrialized landscapes may have lesser contrasts with new structures than if they were positioned on more natural coastlines. Thus, modifications can, instead, mean the final event in a succession of a series of view-polluting developments (Nordman et al., 2015). The same issue affected onshore developments, being the decreasing number of available grounds, and the current legal conflicts around onshore siting often mentioned as central reasons to support the implementation of offshore parks.

After reading the results, no doubts remain about the connection between acceptance and rejection rates, the design of the turbines, and the distance of the OWFs to the shore and communities. The impacts of Offshore Wind Farms reach the tourism sector during the Operation phase (Harvey et al., 2017), while landscape impact is present also during Planning, Construction, and Connection. Throughout this OWF lifecycle the negative impacts, if existent, are likely to come followed by other consequent pressures, as losses on expenditure lead to local loss of income and jobs. Thus, the interaction between time and acceptance is

also a crucial indicator for this impact analysis.

#### 4.1.3. Time as a possible driver and indicator of change

As the results have revealed (Kipperberg et al., 2019; Rydin et al., 2018; Nordman et al., 2016), time is a deciding parameter to any development, regarding legal constraints, developing stages, strategic plans and deadlines, targets, goals, and also for appropriate information dissemination. In this study, time anchors how the analyzed knowledge gap and sectors evolve, how OWF developments affect coastal communities and this multi-use scenario unfolds.

Potential tourism and recreational impacts unravel a discussion that can lead to acceptance and support or to fierce opposition and even motivation for canceling OWF projects, as previously mentioned. Within our results, negative impacts on tourism and displacement of recreational activities are identified as "Controllable" impacts, which means that there is control over the severity of its consequences as they are dependent on manageable factors. However, time limits this control, as developers depend on it to seek constant improvement after the initial plan is set in motion and secure the capital invested. Simultaneously, locals need time to intervene and to provide their opinions as input on the different phases of an OWF park. Nevertheless, the literature shows clear chronological evolutions in public attitudes.

The results showed that populations who lacked information or did not have strong opinions at first slowly gravitated toward more extreme ones. In other words, most became supportive when the parties clarified the multi-use situation. On the other hand, rejection-like answers would be inclined to diminish with time. In general, more informed actors seem to produce louder respondents, people who tend to uphold their opinion and even harden it as time goes by (Rydin et al., 2018). Society's adaptability to changes and the fact that time is likely to clarify significant aspects by increasing the information level among stakeholders can justify this phenomenon.

Decades of land use planning cause people to be more familiar with changes in land use than with the use of marine space. Any change in activities or use of the marine environment seems to represent an unsettling contrast for the population, especially on islanders and traditional coastal communities. The evolution of activities in the marine environment, through the development of the blue economy, is connected to a growing sector still highly dependent on appropriate policies and the implementation of marine spatial planning. Therefore, MSP is a key tool for organizing and reaching a consensus on uses and activities in the marine environment in the same way that land use planning was a crucial step in implementing certain activities (Kerr et al., 2014).

### 4.2. Information, public acceptance, and stakeholders' engagement

The acceptance levels among T&R players affected by installed or planned OWFs and the parameters that affect this acceptance of OWF among coastal inhabitants are observed in our results. The studies converge on a common pattern where younger respondents tend to be more likely to be supportive of disruptive changes in the system and older respondents tend to express opposition to the developments. This behavior lies in the belief that activities such as scuba diving and sport angling are crucial to the continuation of these professions and their communities. They guarantee a safe transition from traditional fishing practices to a new sector where fishing activities also account for leisure fishing. These findings corroborate with others that underline the stakeholders' choice as a central theme in conflict management (Chen et al., 2015).

It is not a surprise that the US, Europe, and Asia dominate the results as it fits the current wind power offshore global capacity profile and matches the positions of Europe, North America, and Asia in the number of offshore parks developed or under development (Kusch-Brandt, 2019). Nonetheless, the US is far behind Europe on offshore wind generation, making the number of papers analyzing the possible impact on tourism and recreation and the number of OWFs installed in these two locations seem incompatible. According to 4C Offshore (4C Offshore, 2020), a specialized market consultancy firm, the US currently has more than 150 offshore wind projects, none of these projects is in the build phase, and just 1 commercial and 1 pilot project currently operating. Europe has hundreds of operational farms, 28 only in Germany, and over a thousand projects. Additionally, Americans install or plan to install OWFs in areas distinguished by widespread visitation and leisure, whereas European projects placed most of their parks in secluded coastal spaces or areas defined by seasonal activities. This fairly surprising result seems also connected to the amount of funding allocated by the U. S. Department of Energy's Wind Energy Technologies Office to related studies and partnerships with universities, as exemplified by Parsons and Firestone (Parsons and Firestone, 2018). This lack of research interest, together with a diversity of cultural factors and business strategies, could be the reason behind this disparity; also influencing how the population participates and influences the decisions based on the local levels of awareness and visibility of the topic. Understanding the discrepancy found in the geographical comparison of publications per OWF developments is regarded as an interesting topic for future research.

Likewise, having a clear public socio-cultural discourse is a critical component for identifying indicators, the clarity of the process, and the perception of fairness and transparency, allowing the community to express their opinions and might play a key role in generating higher support for an OWF project. It reflects and influences social and political concern and (or) support for environmental or government projects (Smith et al., 2018). It helps coastal communities to engage in a fruitful and resilient Blue Economy with the development of nautical routes (natural, cultural, historical), production of tourism branding, promotion of new tourism models, the establishment of artificial reefs, exploration, and enhancement of underwater cultural heritage (Firestone et al., 2018b). Therefore, another valuable conclusion is that opposition rates would decrease if the planning of the offshore wind farms applies a more extensive stakeholders' consultation process. This would require the application of adequate management methods, with standardized compensation claims and more readily available scientific data (Gray et al., 2010), and the participation of smaller stakeholders.

Early-stage conversations suggest that coastal community members understand the benefits and impacts of wind farm development on their quality of life and share complex and specific preferences for the location of offshore turbines, besides having particular definitions for the development of the MRE with other sectors through synergies. Nevertheless, reaching these groups is often a troublesome task. Different methods can break communication barriers and find gatekeepers, such as having local actors genuinely involved with the community as a mediator to make the communication relatable and manageable (Rydin et al., 2018), to show how approximation efforts are notably appreciated by the local stakeholders in this sort of situation. Yet, the authors also criticize the lack of substantial information to prove the concerns.

When it comes to common knowledge, people are generally more informed about the benefits of renewable energy but less so about personal economic costs. After a first contact where they understand that energy developments might directly impact them, individuals are more likely to react less positively (Bayulgen and Benegal, 2019). Here, 92 % of the papers considered the existence of impact and that, if existent, this impact represents mostly undesirable outcomes for tourism and recreation. However, when the analysis comes to the local economic level, the interests and impacts become unclear, and a few authors rank the assumption that these projects will produce local economic benefits as overstated. For example, Rydin et al. (2018) stress the OWF developments using the number of direct jobs generated by the OWF to increase acceptance, as they are likely to be offered to non-local trained experts, a common issue in these small coastal communities. This issue often happens due to the lack of highly skilled workers among the local community members, and the lack of specific measures to keep other jobs within the local sphere. Our findings confirm that these parameters

are often forgotten when companies present job creation numbers, and this must change if the fair transition is to be taken seriously.

Further advances in conversation and openness can solve possible misunderstandings, allowing constructive dialogues, as discussed above when we analyzed the impacts through time. Our results also demonstrate that the behavior toward power production technologies varies even between the different media (editorial publications versus social media) covering it when the discussion reaches the greater public. This perceived result meets the work developed by Nuortimo & Härkönen (2018) and demonstrates another topic worthy of an in-depth study in the future.

In conclusion, a scenario in which Marine Spatial Planning can be implemented and well connected with land planning is possible if the OWF developments consider traditions, local knowledge, $CO_2$ emission reduction, economic opportunities and decent jobs, build strong communication, and produce trustworthy and available data, especially in islands. These characteristics come as a decisive local acceptance or rejection factor in most studies.

#### 4.2.1. Green marketing

The link between green marketing and public acceptance provides a multifaceted starting point for the relationship analyzed here. Tourism and recreational businesses recognize renewables as means to increase their market and international growth. Especially in locations previously suffering from electricity shortages and lack of energy security, such as islands and remote coastal communities, changes in the electricity type and availability can boost tourism and leisure activities by adding new options to their portfolio of activities. These changes are highly connected with external drivers, such as governmental decisions, but can also happen based on internal drivers, such as community-based decisions. T&R local stakeholders often turn into decision-makers themselves by voluntarily changing their local businesses and practices concerning the use of resources, and connecting greener energy with greener tourism (Lucrezi and Saayman, 2017).

The results indicate that presenting the connection between OWFs and cleaner energy, and consequently the pollution reduction, with clear and accessible information is an important way to overcome any possible negative impacts. For example, the literature shows that tourists who hold an ecological view or regularly buy 'green' products are more likely to accept local wind farm developments and less likely to display the NIMBY syndrome (Westerberg et al., 2015). On the other hand, the authors do indicate that this acquired acceptance is dependent and highly varies according to the potential economic, commercial, and recreational losses. Opponents tend to be more conservative, having offshore wind parks "often associated with anti-environment views" (Bates and Firestone, 2015) due to the effects of these developments on multiple local activities. Nevertheless, offshore wind developers are not the only ones to face legislative changes and public pressure.

While not the focus of this study, it is relevant to point out that Tourism and Recreation (T&R) activities can endanger the system and influence the relationship analyzed here. The Maritime Leisure sector can be associated with both eco-tourism and significant anthropogenic emissions, being a relevant climate change driver estimated to contribute to 5% of global anthropogenic Greenhouse gases through transport-related emissions (World Tourism Organization and International Transport Forum, 2019), reinforcing the importance of a detailed and integrative approach. The unplanned growth of hotels and tourism facilities, and the infrastructures created to diminish the traffic congestion on coastal roads with little regard to visual impacts or local architecture, for example, can highly affect the beach's integrity and influence several environmental and cultural aspects. The typical repercussions of inhabitants constantly being outnumbered by the increasing seasonal population are, among others, the induced development pressures aimed at meeting tourist needs, loss of habitats and biodiversity, unsustainable exploitation of natural resources – including excessive abstraction of drinking water, and the consequent lack of sewage treatment and appropriate waste disposal (European Environment Agency, 2015).

Therefore, OWF developers employing the green factor and environmentally friendly associations must go beyond the clean energy production speech. They must investigate, understand and disseminate other potential benefits such as the establishment of fish habitats, artificial reefs, and changes in water quality due to the decrease in pollution and greenhouse gas emissions that can benefit all users of this maritime space.

### 4.3. Tourism and recreation impact assessment (T&R IA)

Offshore wind parks can generate both positive and negative nonmarket impacts (Gudding et al., 2018), and visual deterioration features as the main reason for opposition from local communities. Offshore wind farms can have effects ranging from visitors' behavior and community's revenue to sea-bed morphology changes, sediment transport, species distribution, disturbance through noise and vibration of turbines or the magnetic fields of power cables, the entanglement of marine mammals in mooring lines, or rotating turbines, and collisions of birds with infrastructure above sea-level (United Nations, 2016). In the case of islands, the opposition toward wind farms is higher than wave and tidal technologies, as these other forms of marine energy tend to be less apparent in the seascape, but the use of flashing lights and exclusion zone markers seems to increase the visual impact in all cases (Martino et al., 2019; Bonar et al., 2015; de Groot and Bailey, 2016). On the other hand, less analyzed and understood, however, are some particular effects such as safety issues near turbines, which act indirectly on local tourism-related businesses threatening their ability to get insurance coverage. This business-factor analysis, discussed in more detail by Rydin et al. (2018) and Kermagoret et al. (2016), finds that most concerns and externalities stay at a local level, while most of the revenues and benefits leak toward non-local actors. This fragility seems to be associated with the infrastructure of these projects, shifts in the local economy, and impact on specific businesses.

The World Ocean Assessment highlights that most studies about the possible impacts of tourism activities discuss the increasing growth rate of tourism. Consequently, offshore wind parks could become biodiversity refuges, with the foundations of the devices becoming home for artificial reefs and preventing over-fishing and trawling, as offshore energy farms are likely to constitute prohibited areas for commercial fisheries and navigation. Additionally, the three most common combinations and interactions mentioned under OWF and Coastal and Maritime Tourism were turbine visiting, Pescatourism, and activities associated with the development of Artificial Reefs. These new uses, mentioned by several of the analyzed authors (Martino et al., 2019; Sijmons et al., 2017; Calado et al., 2019; Lin et al., 2019; Westerberg et al., 2015; Klain et al., 2018; Bonnevie et al., 2019), could stop the constant overexploitation of coastal areas, thus helping the regeneration of certain species (United Nations, 2016). Nevertheless, the number and type of multi-use combinations, and exclusion zones, are still inadequately defined and differ around the world. Such synergies need a legal framework specifically addressing the relationship between OWFs and Coastal and Maritime T&R, following the established effect direction of our study, as the tradeoffs discussed seem to affect the tourism sector the most.

The European Union, through legislation, promotes the rise of green infrastructure, or in this case "blue infrastructure". For example, Directive 2014/89/EU on the common framework for the management of maritime space proposes the use of marine environment management for the development of activities, following the Marine Strategy Framework Directive (2008/56/EC). This is the background to the European Union's push for the development of this type of activity, and the relevance of MSP for their implementation. Countries such as the United Kingdom, Germany, and the Netherlands have MSP plans that allowed for greater promotion of OWFs (Quero García et al., 2019). Furthermore,

ad hoc international agreements and meetings on the environment and climate change introduce OWFs to the international scene. Such is the case of COP27, for which a manifesto has been developed by various energy companies for the development and implementation of wind farms (including OWFs) (Global Wind Energy Council (GWEC), 2022). This highlights the relevance of carrying out impact and perception studies of this type of installation, since, on the one hand, the legislation promotes their implementation, but, on the other hand, it could be the perfect environment for an OWF expansion without specific impact studies.

The definition of the parameters for visual evaluation of these impacts has progressed due to the mandatory nature of the Environmental Impact Assessment and its report, Environmental Impact Statement. However, most regulations do not ask for a tourism impact analysis or a tourism impact statement. OWF developers should be mandated to integrate tourism and recreation parameters into the landscape and visual assessment already performed. But our results prove that performing this analysis in detail is crucial to guarantee the balance between the activities of both sectors and lower the possible negative impacts. Thus, we go above and beyond by recommending the development of an appropriate and distinct T&R IA statement. OWF development proposals should define every infrastructural, economic, environmental, and cultural component while defining how these will affect the seascape and landscape. It is essential for producing a proper Tourism and Recreation Impact Assessment, especially in areas where these sectors are the main local source of income, that this document incorporates data from existing or proposed developments and applies the lessons learned while listening to all relevant actors, including both tourists and recreationists. This assessment must be clear and detailed, comprising all development phases, types of impacts, and related mitigatory and preventive practices, conducting several evaluations, and detailed mapping of the area.

Nevertheless, the lack of implemented parks and studies about the interactions described does not allow the confirmation of these apprehensions. It enhances the complexity of the matter and highlights its lack of pattern for generalizations. Scientists and developers have a long way to go before the impacts are well understood and characterized. However, a slow indication of change arises, and efforts to diminish this gap gradually appear as a derivative of the records on OWFs development and the increasing interest in this relation.

## 5. Conclusion

The present analysis proposed a synthesis of papers produced from 2015 to 2019 to answer how the OW sector influences Coastal and Maritime Tourism and Recreation and its possible impacts, evaluating public and scientific awareness of and worries about the subject. The overview discussed the six hypotheses, confirming four and critically assessing the projections over the other two.

The currently available literature depicts the direct and indirect effects of this relationship. The relevant studies show this impact and its perception as predominantly negative, which follows the initial expectations. Negative and positive impacts vary significantly due to local characteristics, OWF management, multi-level policies, Marine Spatial Planning, and successful agreements forged through adequate zoning and integrative coastal management approaches. On the other hand, contrary to the hypotheses, the projected losses are avoidable. The impactful nature of this multi-use of the maritime zone is considerably dependent on how the OWF project is managed and on early stakeholder engagement, on mutually available, transparent, and reliable, information.

Here we raise the need for a detailed tourism and recreation impact assessment that will map the needs and demands of all stakeholders involved to help ensure the ongoing energy transition brings the needed benefits without harming local traditions and these activities. We recommend the Participatory GIS technique and the implementation of on-site interviews for further studies on this topic, to ground this analysis by better quantifying this impact. These extra steps also work as recommendations for developing a Coastal Tourism Impact Assessment model.

## Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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## Data availability

Data will be made available on request.

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Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the institutions involved.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eiar.2022.106999.

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